

# Fabrication of medetomidines/epoxy coatings for marine anticorrosion and antifouling

Miao-Qing Sheng<sup>a</sup>, Yong-Yin Cui<sup>a</sup>, An-Ji Wu<sup>b</sup>, Ming-Xia Song<sup>b</sup>, Ming-Yang Zhang<sup>a</sup>, Yuan Feng<sup>a</sup>, Hui-Jing Li<sup>a</sup>, Yan-Chao Wu<sup>a,\*</sup>

<sup>a</sup> Weihai Marine Organism & Medical Technology Research Institute, College of Marine Science and Technology, Harbin Institute of Technology, Weihai 264209, PR China

<sup>b</sup> Weihai No.1 High School, Weihai 264200, PR China

## ARTICLE INFO

### Keywords:

Medetomidine  
Epoxy coating  
Anticorrosion  
Antifouling

## ABSTRACT

Corrosion and biological fouling present notable challenges for the growth of the marine industry. Since medetomidine (MM) has been shown to induce hyperactivity and interfere with barnacle larvae settlement, we synthesized MM and its analogues, 4-(1-(3,4-dimethyl phenyl)ethyl)-1H-imidazole (MM-2), 5-(1-phenylethyl)-1H-imidazole (MM-3) and 4-(1-(3,4-dichlorophenyl)ethyl)-1H-imidazole (MM-4), to explore their antifouling effectiveness in marine settings. The anticorrosive epoxy resin (EP) was integrated with these antifouling medetomidines to develop coatings that offer both antifouling and anticorrosion benefits. Among the compounds assessed, dichlorosubstituted medetomidine analogue (MM-4) stood out as the most performing candidate. Compared to EP/MM, the inhibition rates of EP/MM-4 against *Escherichia coli* and *Staphylococcus aureus* increased by 24.69 % and 39.28 %, respectively. Moreover, EP/MM-4 showed impressive activities against marine bacteria *Vibrio fischeri* and *Cyclotella cryptica*, with 91.93 % and 91.21 % inhibition rates, while maintaining stability in the coating by the fourth day. Notably, the EP/MM-4 coating not only offers effective antibacterial and algal resistance, but also help prevents large organisms from deposition in seawater for up to 90 d. The impedance value of MM-4 remains at 6500  $\Omega \cdot \text{cm}^2$ , demonstrating excellent anticorrosion performance, with a continuous anticorrosion efficiency of 97.21 % for 24 h. This research holds great promise for practical application in marine and industrial facilities has broad prospects.

## 1. Introduction

Marine biological fouling refers to the undesirable attachment of various organisms, including bacteria, algae, and barnacles, to wet surfaces [1,2]. Marine surfaces such as offshore drilling platforms, transportation pipelines and vessels that are submerged in seawater for prolonged periods are particularly vulnerable to colonization by marine organisms [3–5]. The fouling process is a gradual in nature [6]. Initially, bacteria adhere to the surface of the coating and proteinaceous substances that facilitate the formation of a “biofilm” conducive to further biological proliferation [7]. Subsequently, organisms such as algae and plankton attach themselves to this biofilm layer, resulting in changes to the composition of the membrane surface over time. In summary, the presence of biological parasites leads to a gradual increase in pollution levels. The biofilm formation is a critical step in subsequent biological fouling and may also worsen surface corrosion [8,9]. When organic

matter accumulates on surface, it causes localized changes in ion types and concentrations, oxygen content, and pH value [10,11]. This process can promote the coatings biodegradation, increase liquid conductivity, and encourage both chemical and electrochemical reactions [12]. This corrosion type is famously as microbiological-influenced-corrosion [13–15]. Additionally, seawater has a salinity of about 3.5 % [16], making it a highly corrosive medium that can trigger chemical and electrochemical reactions at the seafloor surface. Marine corrosion is capable of creating small cracks on surface that gradually lead to more extensive deterioration, compromising structural integrity and presenting serious safety concerns [17,18]. Therefore, developing effective marine anticorrosion and antifouling coatings has become an important priority [19–21].

Medetomidine (MM) is a commercially available medication known for its sedative properties [22,23] and serves as an environmentally friendly antifoulant [24] that effectively prevents barnacles and other

\* Corresponding author.

E-mail address: [ycwu@iccas.ac.cn](mailto:ycwu@iccas.ac.cn) (Y.-C. Wu).

<https://doi.org/10.1016/j.porgcoat.2024.109033>

Received 31 October 2024; Received in revised form 16 December 2024; Accepted 20 December 2024

Available online 2 January 2025

0300-9440/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

marine organisms from settling on the hull surfaces, significantly reducing navigational resistance while demonstrating commendable antifouling efficacy [25,26]. L. Ulrika reported that MM can induce hyperactivity in barnacles, thereby disrupting their settlement process and achieving notable antibiofouling effects [27]. One approach to enhance antifouling performance involves the synthesis of MM analogues followed by structural-activity relationship studies. Moreover, antifouling medetomidines hold promise for integration with anticorrosion epoxy resin (EP), facilitating the development of composite coatings that offer both antifouling and anticorrosion benefits. EP exhibits strong adhesion to various substrates along with excellent corrosion resistance [28], however, it lacks inherent antifouling characteristics, making it susceptible to microbiologically influenced corrosion [13–15] which may compromise its anticorrosive effectiveness. Incorporating an antifoulant into EP coating would mitigate the attachment of marine organisms and reduce microbiologically influenced corrosion risks [29]. M.S. Noor Idora [30] investigated the antifouling properties of EP through addition of varying concentrations of powder extracts, revealing that a low concentration of zinc tannate (2.736 mg/mL), was adequate to inhibit bacterial growth in *Bacillus cereus*.

In this study, antifouling MM and its representative analogues, 4-(1-(3,4-dimethyl phenyl)ethyl)-1H-imidazole (MM-2), 5-(1-phenylethyl)-1H-imidazole (MM-3) and 4-(1-(3,4-dichlorophenyl)ethyl)-1H-imidazole (MM-4), are synthesized from readily available starting materials. When combined with anticorrosion EP, they serve as probes to assess the potential of medetomidines/epoxy coatings for marine anticorrosion and antifouling applications (Fig. 1). The composite EP/MM-4 coating not only demonstrates exceptional marine anticorrosion performance but also effectively prevent the attachment of marine organisms to the hull. The antibacterial activities of these composite coatings were evaluated applying *Staphylococcus aureus* (*S. aureus*), *Escherichia coli*

(*E. coli*) and the marine bacteria *Vibrio fischeri* (*V. fischeri*), while their anti-algal properties were investigated using *Chlorella cryptica* (*C. cryptica*). The synthesized MM-type antifouling agents also exhibit excellent anticorrosion performance. Additionally, the slow-release rates of antifoulants from the composite coatings were measured, and both antifouling and anticorrosion performances were tested in seawater over a period of to 90 d.

## 2. Experimental section

### 2.1. Materials

2,3-Dimethyl acetophenone (99.5 %) and 1-(trimethylsilyl)-1H-imidazole ( $\geq 95.0$  %) were obtained from Shanghai Haohong Biological Medicine Technology Co., Ltd., while sodium borohydride ( $\text{NaBH}_4$ , AR) was sourced from Tianjin Comeio Chemical Reagent Co., Ltd. Trichloromethane (AR) was supplied by Yantai Far East Fine Chemical Co., Ltd., and hydrochloric acid (HCl, AR) was procured from Yantai Shuangshuang Chemical Co., Ltd. Titanium tetrachloride (99 %) was provided by Annaji Chemical & 3A, whereas bisphenol A diglycidyl ether (E51, epoxy equivalent: 182–192 g/mol), thus polyether amine curing agent styled a D230, as Mn = 230 g/mol and defoamer were acquired from Shandong Maifei Chemical Co., Ltd.

### 2.2. Synthesis of medetomidines

To a solution of 2,3-dimethylacetophenone (2 mmol) in anhydrous methanol was added sodium borohydride (4 mmol) at 0 °C. The resulting mixture was stirred at 0 °C for 12 h, and quenched by distilled water. The aqueous mixture was extracted with methylene chloride (20 mL  $\times$  3). The combined organic extracts were dried over anhydrous sodium sulfate, filtered, and concentrated. The residue was subjected to

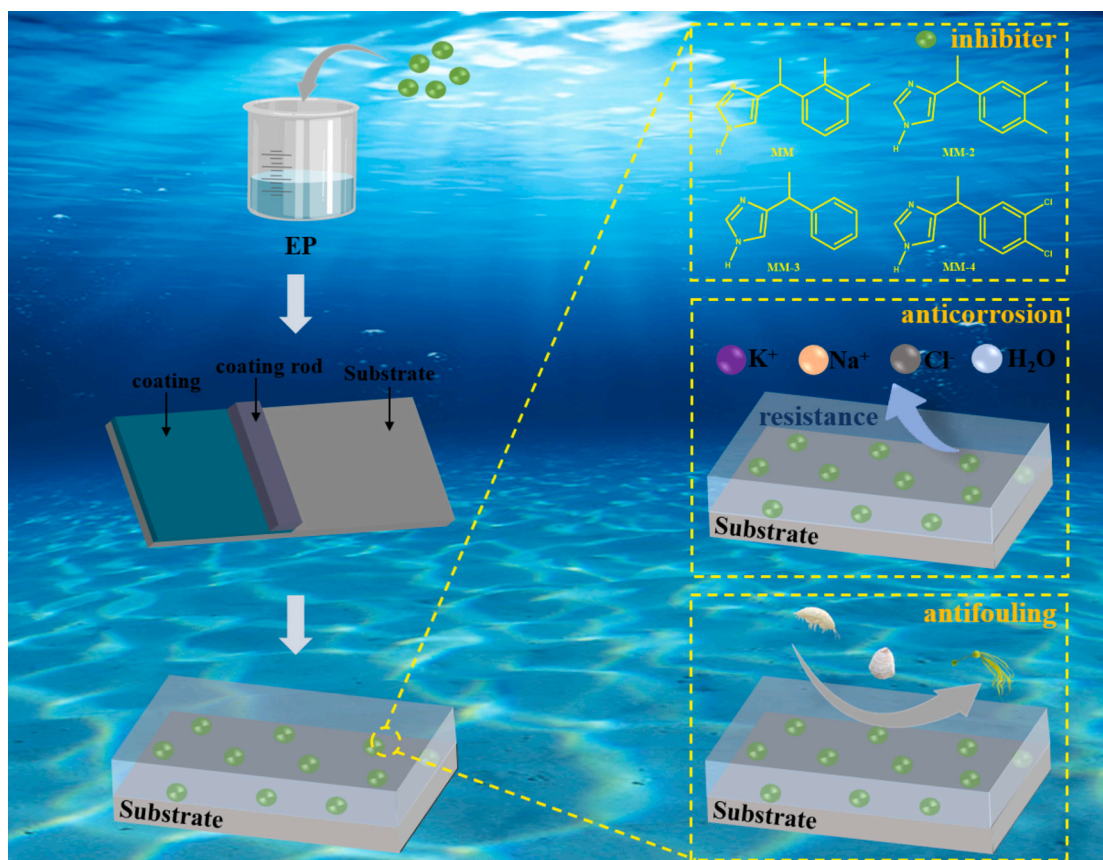


Fig. 1. Molecular design of antifouling and anticorrosion coatings.

flash column chromatography (PE:EA = 30:1) on silica gel to afford 1-(2,3-dimethylphenyl) ethanol in 95 % yield.

To a solution of trimethylsilylimidazole (1.0 equiv.) in chloroform (15 mL) under the atmosphere of nitrogen was added titanium tetrachloride (1.0 equiv.). The resulting mixture was stirred at room temperature for 2 h, added with 1-(2,3-dimethylphenyl) ethanol (0.5 equiv.), and then stirred at room temperature for about 6 h. The reaction time should be long enough to ensure the full reaction of 1-(2,3-dimethyl phenyl) ethanol, and the yield of MM is 75 %. MM analogues (MM-2, MM-3, MM-4) are also synthesized using this protocol, and the synthesis route is shown in Fig. 2.

### 2.3. Coating preparation

A blend of E51 and D230 in a ratio of 100/31 was thoroughly mixed at 40 °C. As this mixture MM (3 wt%) and defoamer (1 wt%) were incorporated. Following 15 min of ultrasonic stirring, the mixture underwent bubble removal under vacuum for 5 min. It was then poured into a Teflon mold and at 60 °C curing for 12 h prior to testing. The resulting mixture was applied to an epoxy plate with a thickness of 30  $\mu\text{m}$ , using the same curing conditions, referred to as EP/MM.

### 2.4. Characterization methods

#### 2.4.1. $^1\text{H}$ nuclear magnetic resonance ( $^1\text{H}$ NMR)

At room temperature and in  $\text{CDCl}_3$  solvent,  $^1\text{H}$  NMR spectra were recorded on the ADVANCE III 400 MHz NMR spectrometer of the German Bruker company.

#### 2.4.2. Anticorrosion test

Comprehensive characterization methods are detailed in SI.

#### 2.4.3. Adhesion examination

Comprehensive characterization methods are detailed in SI.

#### 2.4.4. Antifouling test

Comprehensive characterization methods are detailed in SI.

## 3. Results and discussion

### 3.1. Synthesis of medetomidine analogues

The stability of corrosion inhibitor molecules' adsorption on metal surfaces was mainly influenced by their structure and properties. Heteroatoms could donate electrons to the vacant d orbital of transition metal, forming coordination covalent bonds the process. Additionally, delocalized  $\pi$  electron systems could accept free electrons from the surface of transition metal, creating a feedback bond [31]. Nitrogen-containing heterocycles lone their pair electrons, a conjugated structure, and electronegative structure [17,31] that allowed them to be polycentric adsorbed on the metal surfaces, acting as components in

various metal corrosion inhibitors. The antifouling compound MM and its representative analogues (MM-2, MM-3, MM-4) (Fig. 3) feature nitrogen-rich and aromatic rings that align with essential criteria for effective corrosion inhibitors. These four MM-type compounds have been successfully designed and synthesized, and their  $^1\text{H}$  NMR spectra are shown in Fig. 3.

### 3.2. Anticorrosion property of medetomidine analogues

These findings from the polarization curve analysis were shown in Fig. 4a. The curve was utilized to accurately delineate the mechanisms underlying anode and cathode corrosion reaction, as well as to further investigate the corrosion properties of MM and its analogues. The polarization parameters presented in Table 1 were processed using the extrapolation way of linear Tafel curves [32]. The corrosion potential ( $E_{\text{corr}}$ ) of carbon steel in seawater is  $-611$  mV. After adding MM and its analogues, the  $E_{\text{corr}}$  moved in the direction of negative potential. The results indicated that MM and its analogues were mixed corrosion inhibitors based on cathode [33,34]. The data reveal that the cathodic branches display low current densities when inhibitors were presented, suggesting that the cathodic reaction could effectively be delayed in the seawater [35]. The observation confirmed that MM and its analogues impeded the hydrogen evolution at cathode interface. Among the evaluated medetomidines, dichlorosubstituted medetomidine analogue (MM-4) demonstrated exceptional anticorrosion performance at a molar concentration of 400 mg/L, it achieved 94.38 % corrosion inhibition efficiency of MM-4.

EIS measurements could be used to predict the electrochemical reaction process at the interface of the steel-solution, which gave the representative Nyquist and Bode plots (Fig. 4) of MM and its analogues. From Fig. 4b, the Nyquist diagram presents a single semicircle rather than a complete semicircle. This is caused by the decrease in the interface impedance frequency of Q235 steel and the surface roughness and non-uniformity [36]. Consequently, a constant phase element (CPE) replaced the capacitive element ( $C_d$ ) to address inaccuracies. Additionally, introducing MM-type compounds resulted in an increased diameter of capacitor loop [37]. The capacitance loop diameter for MM-4 is significantly larger, with an impedance value of  $6500 \Omega \cdot \text{cm}^2$ . This indicated that MM and its analogues inhibit corrosion of Q235 metal primarily by slowing charge transfer and site reactions. The Bode diagram (Fig. 4c, d) showed that adding MM-type compounds increases both impedance Bode modulus and phase Angle, further clarifying their inhibitory effect on the corrosion.

The EIS data were analyzed with equivalent circuits shown in Fig. 5. The  $R_p$  value ( $R_p = R_{\text{ct}} + R_f$ ) in the presence of a corrosion inhibitor is significantly higher than that without one, especially for MM-4 [38,39]. Specifically, it increased from  $221 \Omega \cdot \text{cm}^2$  to  $7911.1 \Omega \cdot \text{cm}^2$ , yielding a corrosion inhibition efficiency reaches 97.21 % (Table 2). At concentrations of MM, MM-2 and MM-3 were 400 mg/L, the  $R_p$  values rose to  $2855.3 \Omega \cdot \text{cm}^2$ ,  $3361.7 \Omega \cdot \text{cm}^2$  and  $2560.1 \Omega \cdot \text{cm}^2$ , their corresponding inhibition efficiencies were 92.26 %, 93.42 % and 91.37 %, respectively.

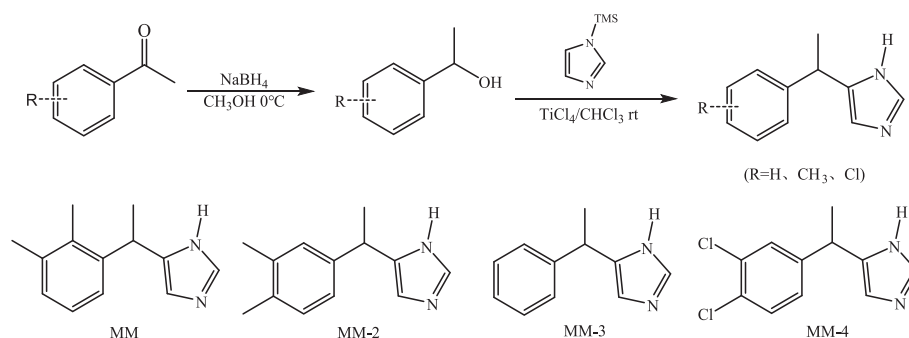


Fig. 2. Synthesis route to medetomidines.

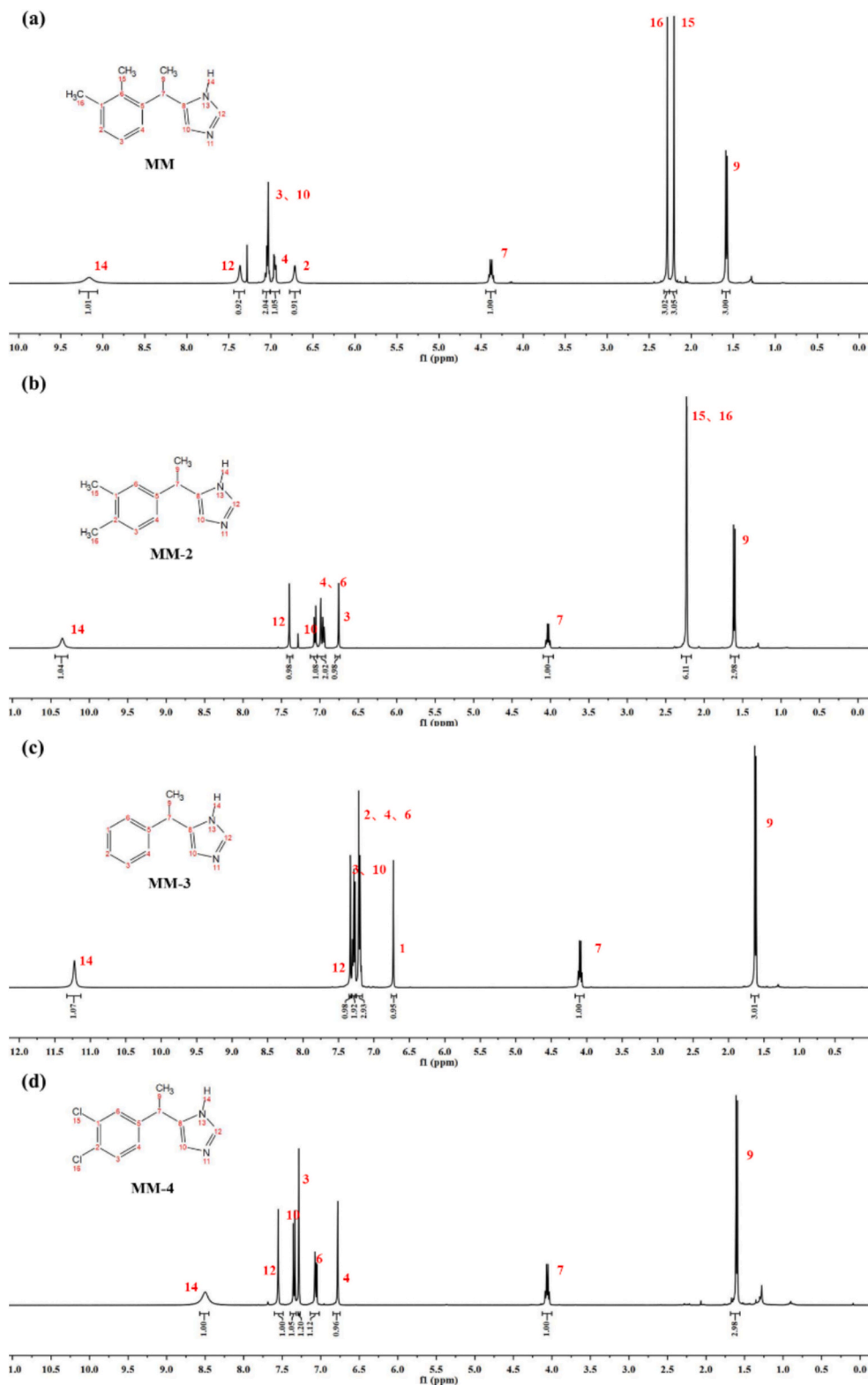


Fig. 3. <sup>1</sup>H NMR spectra of (a) MM, (b) MM-2, (c) MM-3 and (d) MM-4.

These results on metal corrosion inhibition were consistent with those from polarization curve.

As shown in Fig. 6a, SEM was employed to analyze the surface morphology of Q235 steel at 298 K temperature. In an uninhibited solution, the carbon steel surface was severely corroded and had obvious deep pits and an uneven surface. Notably, the metal surface in seawater with MM-type corrosion inhibitors was smoother than in a blank

seawater. The results suggested that MM and its analogues formed a strong film on the Q235 metal surface, reducing seawater corrosion. With MM-4 as the inhibitor, the carbon steel surface was much flatter than with other MM-type corrosion inhibitors, showing the best anti-corrosion performance. The SEM analysis of surface corrosion morphology was consistent with the electrochemical experiment results.

The weight loss test was a valuable method for understanding how

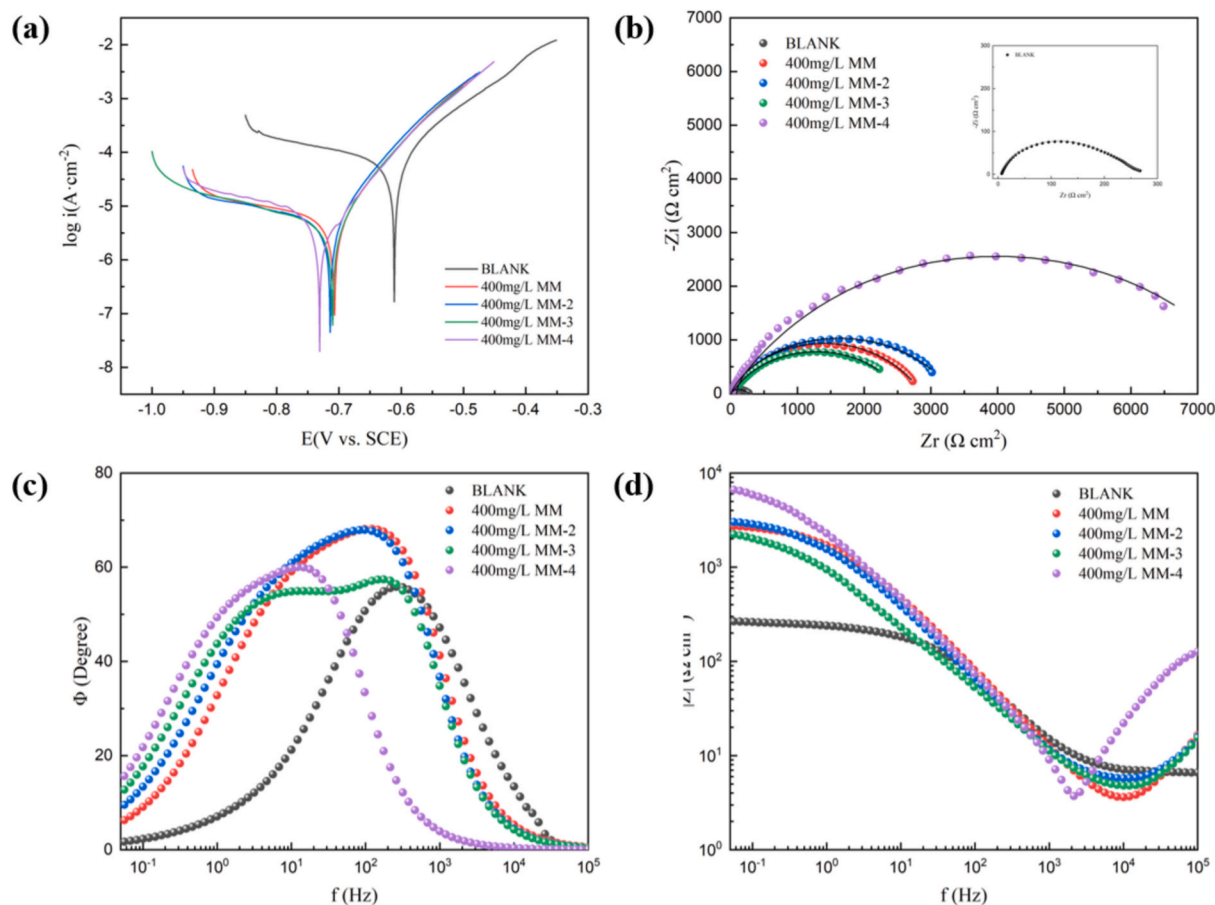


Fig. 4. Different corrosion inhibitors in seawater of (a) Polarization curves, the diagram of (b) Nyquist and (c) (d) Bode.

Table 1

Potentiodynamic polarization curve parameters of carbon steel with/without corrosion inhibitors in seawater.

| Inhibitors | C (mg/L) | $E_{corr}$ (mV) | $i_{corr}$ ( $\mu\text{A cm}^{-2}$ ) | $\beta_a$ (mV $\text{dec}^{-1}$ ) | $-\beta_c$ (mV $\text{dec}^{-1}$ ) | $\eta$ (%) |
|------------|----------|-----------------|--------------------------------------|-----------------------------------|------------------------------------|------------|
| Blank      | –        | –611            | 99.22                                | 139.3                             | 248.2                              | –          |
| MM         | 400      | –707            | 6.73                                 | 117.3                             | 158.6                              | 93.22      |
| MM-2       | 400      | –714            | 6.30                                 | 103.5                             | 179.6                              | 93.65      |
| MM-3       | 400      | –710            | 6.36                                 | 89.5                              | 132.9                              | 93.59      |
| MM-4       | 400      | –731            | 5.57                                 | 88.7                              | 112.5                              | 94.38      |

various temperatures affect the corrosion inhibitor MM and its analogues. The corrosion rate ( $v_{corr}$ ) and corrosion inhibition efficiency ( $\eta$ ) were summarized as Table S2 presented. Before adding the inhibitor, the corrosion rate was recorded at  $5.361 \text{ mg}\cdot\text{cm}^{-2} \text{ h}^{-1}$  (Fig. 6b). With the addition of MM-type corrosion inhibitors, both the corrosion rate decreased steadily and the inhibition efficiency improved as well. These inhibitors created a protective film on the carbon steel surface, helping to prevent erosion from corrosive seawater media. Interestingly, MM showed sustained release efficiency of 86.27 %. At a concentration of 400 mg/L for MM-4, the corrosion inhibition efficiency could reach an impressive 97.05 %, demonstrating excellent anticorrosion performance. It's clear that different substituents within MM-type inhibitors have a significant impact on their effectiveness in inhibiting carbon steel corrosion. Among them, MM-4 stood out due to strong adsorption performance linked to chlorine atoms on carbon steel surface which facilitate coordination bond formation between MM-4 and the metal surface. This interaction improves the charge transfer at carbon steel the interface, leading to enhanced corrosion inhibition effect overall. The results

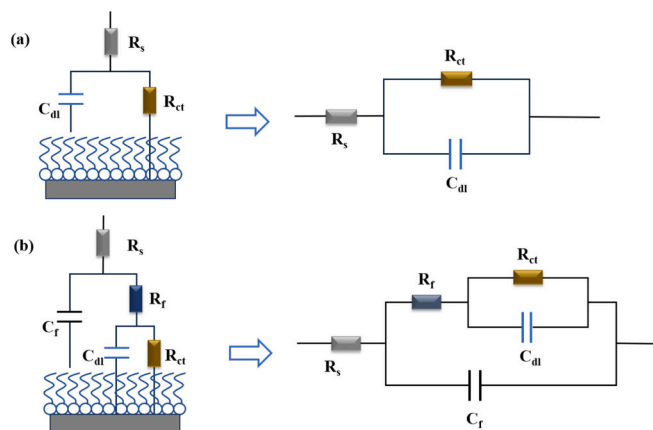


Fig. 5. Equivalent circuit models for Q235 steel in seawater: (a) without and (b) with various corrosion inhibitors.

from the weight loss test were consistent with finding from electrochemical experiments and morphological characterization, further confirming that MM-4 was an effective choice for protecting carbon steel in seawater.

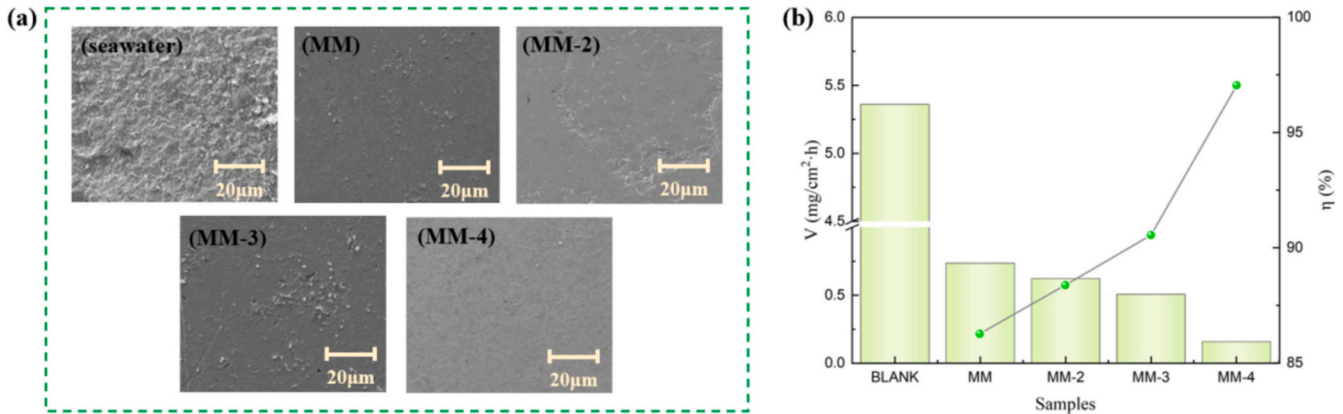
### 3.3. Anticorrosion property of medetomidines/epoxy coatings

As a profound understanding acquired of the coatings' corrosion mechanism, EIS was utilized to examine the selected coatings within a NaCl solution as 3.5 wt%. Fig. 7a illustrated the Nyquist diagram for the

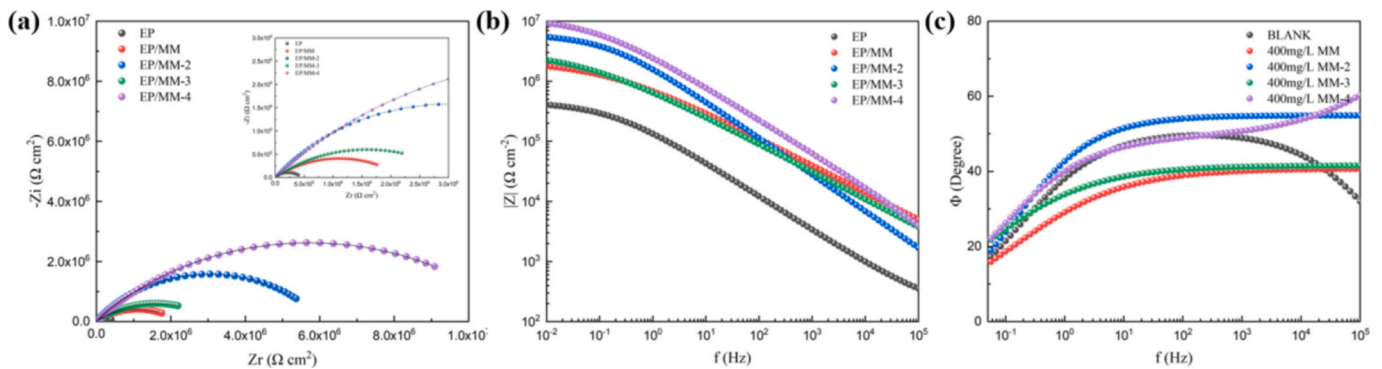
**Table 2**

Impedance parameters of seawater containing corrosion inhibitors.

| Inhibitors | C (mM) | $R_s$ ( $\Omega\text{-cm}^2$ ) | $R_f$ ( $\Omega\text{-cm}^2$ ) | $R_{ct}$ ( $\Omega\text{-cm}^2$ ) | $C_f$ ( $\mu\text{F cm}^{-2}$ ) | $C_{dl}(n)$ ( $\mu\text{F cm}^{-2}$ ) | $R_p$ ( $\Omega\text{-cm}^2$ ) | $\eta$ (%) |
|------------|--------|--------------------------------|--------------------------------|-----------------------------------|---------------------------------|---------------------------------------|--------------------------------|------------|
| Blank      | –      | 8.634                          | –                              | 221                               | –                               | 255.5 (0.52)                          | 221                            | –          |
| MM         | 400    | 10.72                          | 192.3                          | 2663                              | 16.21                           | 89.8 (0.67)                           | 2855.3                         | 92.26      |
| MM-2       | 400    | 9.926                          | 133.7                          | 3228                              | 20.49                           | 121.5 (0.64)                          | 3361.7                         | 93.42      |
| MM-3       | 400    | 9.609                          | 74.1                           | 2485                              | 21.94                           | 230.6 (0.67)                          | 2560.1                         | 91.37      |
| MM-4       | 400    | 9.033                          | 779.1                          | 7132                              | 25.58                           | 95.5 (0.67)                           | 7911.1                         | 97.21      |



**Fig. 6.** (a) SEM images of carbon steel surface under different corrosion inhibitors, and (b) Corrosion efficiency of different corrosion inhibitors in seawater.



**Fig. 7.** EIS test results of the five samples, the curves of (a) EIS Nyquist, (b) Bode plots and (c) Phase angle.

chosen coatings. When contrasted with the EP sample, the other specimens manifested larger capacitive arcs, especially EP/MM-4, signifying that the charge transfer process on the surface of these samples was conspicuously hindered. Generally, a larger the impedance of the sample, and consequently, a greater its corrosion resistance [40]. In Fig. 7b, the Bode diagram further expounded the corrosion behavior of the diverse samples. All the other samples demonstrated higher impedance modulus within the low frequency range in comparison to EP. The impedance modulus of the coating with the inhibitor incorporated was approximately an order of magnitude higher than that of EP. The impedance modulus value within the low frequency region of the Bode chart was frequently employed as an indicator to appraise the corrosion resistance. Generally, a better low-frequency impedance modulus indicated that the coating possessed favorable barrier properties and corrosion resistance. This implied that EP/MM-4 samples showcase the optimum corrosion resistance. The EIS data was subjected to further analysis by means of equivalent circuit fitting (Fig. 5), and the pertinent fitting parameters were summarized in Table S3 within the equivalent circuit. The  $R_{ct}$  of EP/MM-4 coatings was nearly two orders of magnitude higher than that of EP, suggesting that EP/MM-4 coatings notably inhibit the charge transfer process, thereby

significantly enhancing the corrosion resistance. Additionally, the formation of an air cushion on the coating surface reduced the contact between the corrosive medium and the substrate, efficaciously decelerating the corrosion process. In summary, the experimental outcomes obtained in the simulated corrosion environment validate that the sample exhibited remarkable corrosion resistance. Therefore, MM-4 was an outstanding corrosion inhibitor.

### 3.4. Antifouling performance of medetomidines/epoxy coatings

The adhesion grades of composite coatings to the matrix were rated at 5 B, meeting the established adhesion standards (Fig. S3). The addition of MM and its analogues did not negatively impact the coating's adhesion to the substrate. We used Ultraviolet (UV) spectroscopic analysis to evaluate the release rate of MM-type compounds in EP coatings (Fig. 8a). The release rate of medetomidine becomes relatively stable after four days. In addition, the release rate of EP/MM-4 began to stabilize at around  $2.1 \mu\text{g}/(\text{cm}^2\cdot\text{d})$ , showing a higher cumulative release compared to EP/MM. The protein adsorption capacity of the EP coating was measured at  $340 \mu\text{g}/\text{cm}^2$ . However, adding MM significantly boosts this capacity (Fig. 8b), while coating with MM-4 showed only  $198 \mu\text{g}/$

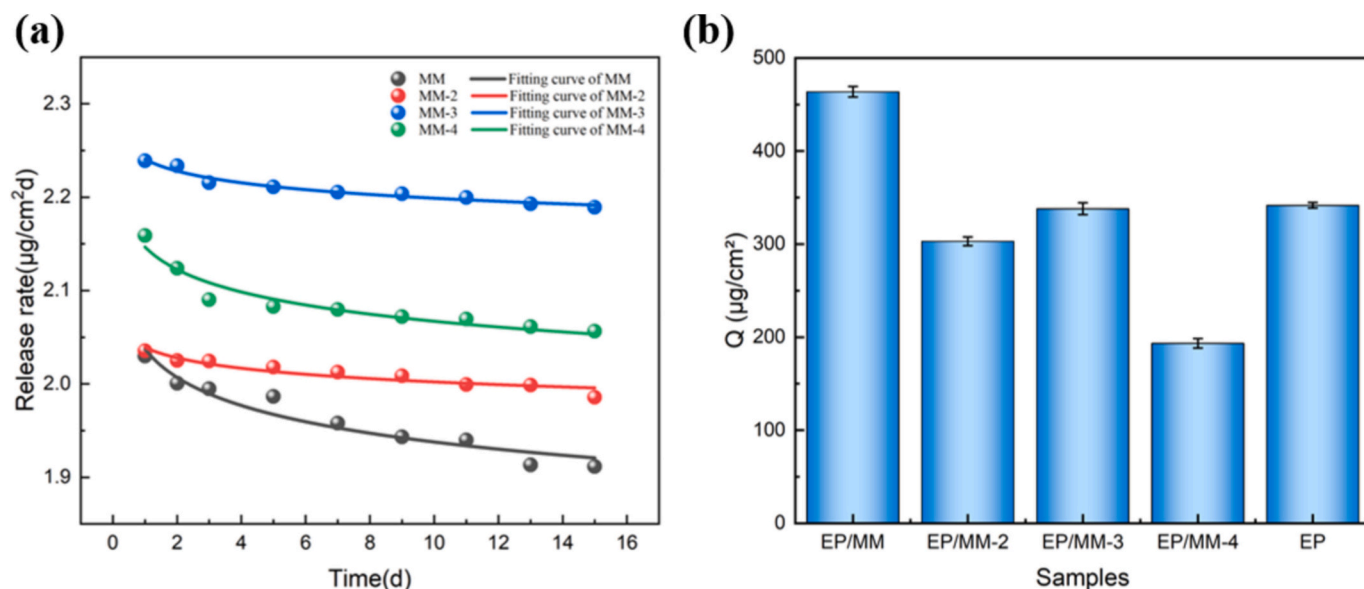


Fig. 8. (a) Release rate of MM and its analogues over time, and (b) Static protein adsorption capacity of the coating.

cm<sup>2</sup>. Reducing protein adsorption on these coatings can effectively slow down biofilm formation, helping achieve antifouling effect. Additionally, stable release rate of antifoulants help inhibit bacterial and algal adhesion on the coating surface.

The antibacterial performance of EP coatings was studied by the flat plate coating method. Tests for *E. coli* and *S. aureus*, representing Gram-negative and Gram-positive bacteria, and the marine bacterium *V. fischeri*, respectively. The antibacterial performance of the coatings can be reflected by the number of bacterial colonies on the plate (Fig. 9a). Compared with EP coating, the antibacterial performance of the coatings with MM and its analogues has been greatly improved. MM and its analogues exhibit species-specific antibacterial properties (Fig. 9b). The antibacterial performance of EP/MM-4 against *S. aureus* is

better than that against *E. coli*. The antibacterial activity of EP/MM-4 against *S. aureus* is 91.08 %, and the antibacterial activity of EP/MM-4 against *V. Fischer* reaches 91.93 %, indicating that EP/MM-4 has good resistance to marine bacteria and Gram-positive bacteria.

The silicon-based *C. cryptica* was selected as the experimental alga. The inhibitory impacts of EP/MM, EP/MM-2, EP/MM-3, EP/MM-4 and PE on algal were observed microscopically. A considerable and uniformly distributed *Cryptococcus* was detected on the EP coating surface (Fig. 9a). In contradistinction to the EP coating, which with MM and its analogues manifested significantly enhanced resistance to algae, with EP/MM-4 showing a remarkable algal resistance of 91.21 % against *Cryptococcus* (Fig. 9c).

Marine equipment confronts the dual challenges of antifouling and

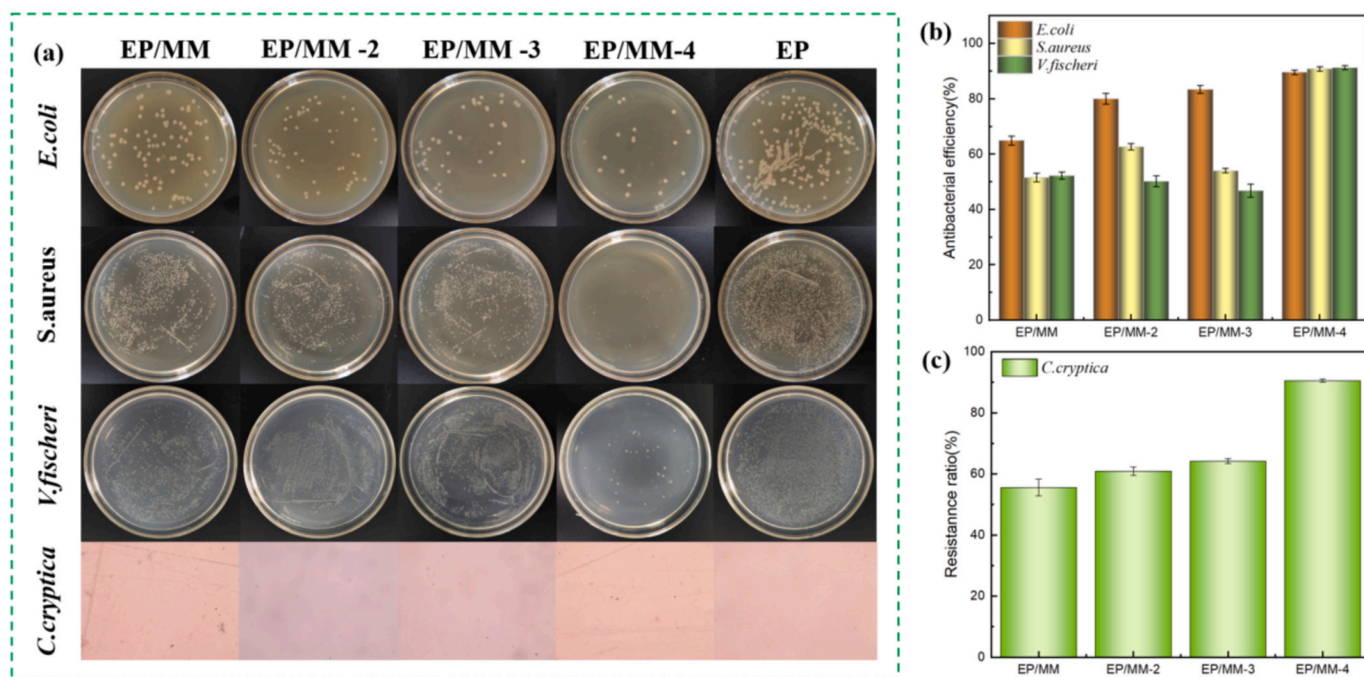


Fig. 9. Antibacterial and algal inhibitory properties of EP, EP/MM, EP/MM-2, EP/MM-3 and EP/MM-4. (a) Plate count images of *E. coli*, *S. aureus* and *V. fischeri* sample, along with digital micrographs of *Chlorella* on the coating. (b) Antimicrobial rate against *E. coli*, *S. aureus* and *V. fischeri*; (c) Percentage of antialgal activity against *C. cryptica*.

anticorrosion. As further the antifouling coat performance assessed directly, offshore field evaluation was conducted in Xiaoshidao Bay, Weihai City, China, on May 1st, 2024. Fig. 10 indicates the test panel after immersion in seawater for 0, 15, 30, 45, 60, and 90 d. At 90 d, the EP coating was covered with a large amount of seaweed and other polluting organisms, indicating high pollution pressure at the testing site. MM and its analogues have antifouling properties, enhancing the limited performance of EP coatings through their sustained release. After 90 d, compared with EP coating, the coating with a MM-type antifoulant has notable antifouling ability. Among them, the EP/MM-4 coating has the best antifouling performance and is covered by only a small amount of marine organism. The antifouling performance of EP/MM coating ranks second. In the marine environment, the antifouling performance can prevent further damage to coatings caused by biological pollution. The antifouling modification of EP can be achieved without compromising its anticorrosion ability.

#### 4. Conclusion

In conclusion, MM and its analogues have been designed and successfully synthesized, which and EP were used to prepare composite medetomidines/epoxy coatings. These coatings demonstrate excellent anticorrosion and antifouling properties. The chlorine atoms in MM-4 show improved adsorption properties for carbon steel, which helps form coordination bonds with it, enhancing charge transfer on the steel surface. In seawater, the impedance value for 400 mg/L MM-4 remains

stable at about  $6500 \Omega \cdot \text{cm}^2$ , achieving a corrosion inhibition efficiency of 97.21 %. Adding a corrosion inhibitor to EP coating continues to provide effective resistance against corrosion. In addition, the gradual release of MM-4 in EP/MM-4 can meet the long-term protection needs of offshore equipment, and the release volume becomes relatively stable at about  $2.1 \mu\text{g}/(\text{cm}^2 \cdot \text{d})$  on the fourth day. The inhibition rates of MM-4 against Gram-negative bacteria (*E. coli*) and Gram-positive bacteria (*S. aureus*) were recorded at 90.43 % and 91.08 %. Furthermore, it achieved an impressive inhibition rate of 91.93 % against marine bacteria (*V. fischeri*), demonstrating strong antifouling performance along with a coverage inhibition rate against of 91.21 %. Importantly, after being immersed in seawater for 90 d, the EP/MM-4 coating showed minimal attachment by marine organism. This study presents a simple and environmentally friendly strategy for developing marine antifouling and anticorrosion coatings.

#### CRediT authorship contribution statement

**Miao-Qing Sheng:** Writing – original draft, Investigation. **Yong-Yin Cui:** Writing – original draft, Investigation. **An-Ji Wu:** Validation, Data curation. **Ming-Xia Song:** Writing – review & editing, Supervision. **Ming-Yang Zhang:** Validation, Data curation. **Yuan Feng:** Validation, Data curation. **Hui-Jing Li:** Writing – review & editing, Supervision. **Yan-Chao Wu:** Writing – review & editing, Supervision, Conceptualization.

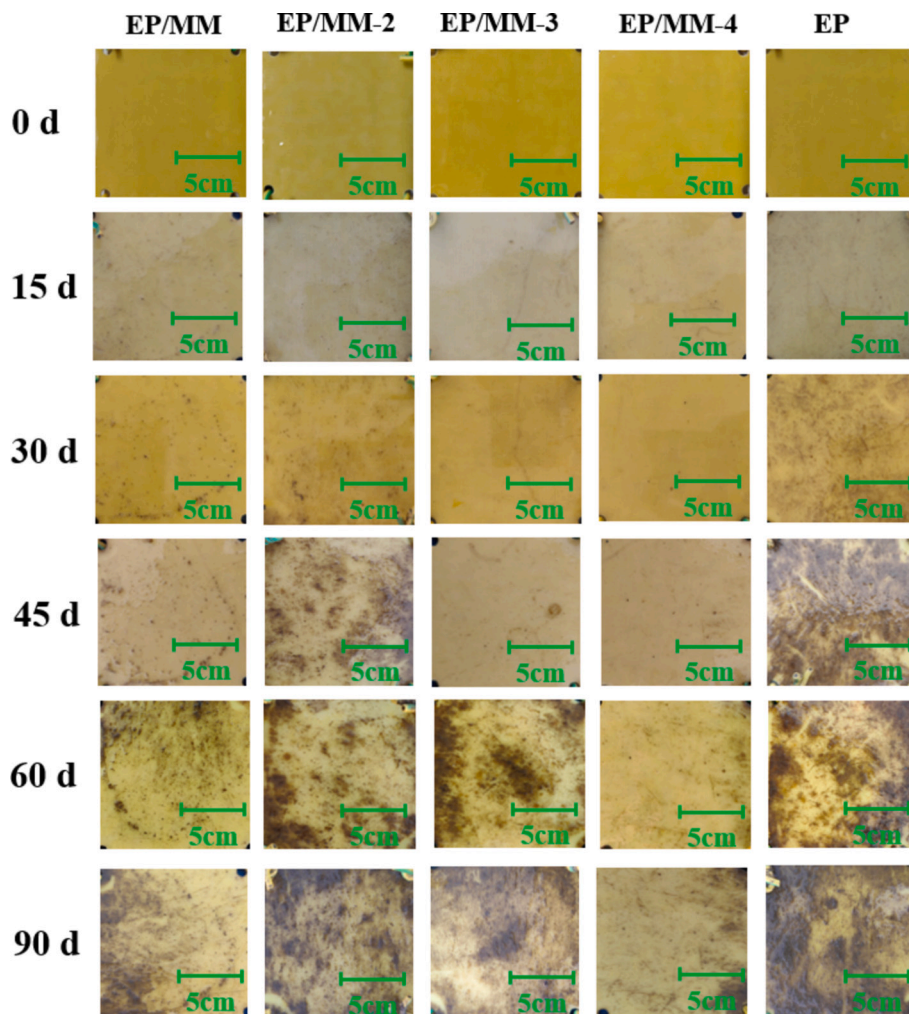


Fig. 10. Images of the coated test panels of EP/MM, EP/MM-2, EP/MM-3, EP/MM-4 and after different days of immersion in seawater.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgment

This work was funded by the National Natural Science Foundation of China (No. 22271070).

## Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.porgcoat.2024.109033>.

## Data availability

Data will be made available on request.

## References

- [1] S. Li, K. Feng, J. Li, Y. Li, Z. Li, L. Yu, X. Xu, Marine antifouling strategies: emerging opportunities for seawater resource utilization, *Chem. Eng. J.* 486 (2024) 149859, <https://doi.org/10.1016/j.cej.2024.149859>.
- [2] F. Nobuhiro, Biofouling and antifouling, *Nat. Prod. Rep.* 21 (2004) 94–104, <https://doi.org/10.1039/b302231p>.
- [3] M.J. Romeu, F. Mergulhão, Development of antifouling strategies for marine applications, *Microorganisms* 11 (2023) 1568, <https://doi.org/10.3390/microorganisms11061568>.
- [4] Z. Cao, P. Cao, Research progress on low-surface-energy antifouling coatings for ship hulls: a review, *Biomimetics* 8 (2023) 502, <https://doi.org/10.3390/biomimetics8060502>.
- [5] T. Gnanasampanthan, J.F. Karthäuser, S. Spöllmann, R. Wanka, H.W. Becker, A. Rosenhahn, Amphiphilic alginate-based layer-by-layer coatings exhibiting resistance against nonspecific protein adsorption and marine biofouling, *ACS Appl. Mater. Interfaces* 14 (2022) 16062–16073, <https://doi.org/10.1021/acsami.2c01809>.
- [6] P. Halder, N. Hossain, B.K. Pramanik, M.A. Bhuiyan, Engineered topographies and hydrodynamics in relation to biofouling control—a review, *Environ. Sci. Pollut. Res. Int.* 28 (2020) 40678–40692, <https://doi.org/10.1007/s11356-020-10864-3>.
- [7] T.T. More, J.S.S. Yadav, S. Yan, R.D. Tyagi, R.Y. Surampalli, Extracellular polymeric substances of bacteria and their potential environmental applications, *J. Environ. Manage.* 144 (2014) 1–25, <https://doi.org/10.1016/j.jenvman.2014.05.010>.
- [8] W. Ma, M. Panecka, N. Tufenkji, M.S. Rahaman, Bacteriophage-based strategies for biofouling control in ultrafiltration: in situ biofouling mitigation, biocidal additives and biofilm cleanser, *J. Colloid Interf. Sci.* 523 (2018) 254–265, <https://doi.org/10.1016/j.jcis.2018.03.105>.
- [9] M. Adnan, E. Alshammari, M. Patel, S.A. Amir, S. Khan, S. Hadi, Significance and potential of marine microbial natural bioactive compounds against biofilms/biofouling: necessity for green chemistry, *Peer J* 6 (2018) 5049, <https://doi.org/10.7717/peerj.5049>.
- [10] A.V. Héctor, Prevention and control of biocorrosion, *Int. Biodeter. Biodegr.* 49 (2002) 259–270, [https://doi.org/10.1016/S0964-8305\(02\)00053-7](https://doi.org/10.1016/S0964-8305(02)00053-7).
- [11] T. Gu, D. Wang, Y. Lekbach, D. Xu, Extracellular electron transfer in microbial biocorrosion, *Curr. Opin. Electrochem.* 29 (2021) 100763, <https://doi.org/10.1016/j.coelec.2021.100763>.
- [12] E. Puentes-Cala, V. Tapia-Perdomo, D. Espinosa-Valbuena, M. Reyes-Reyes, D. Quintero-Santander, S. Vasquez-Dallos, H. Salazar, P. Santamaría-Galvis, R. Silva-Rodríguez, G. Castillo-Villamizar, Microbiologically influenced corrosion: the gap in the field, *Front. Env. Sci.* 10 (2022) 924842, <https://doi.org/10.3389/fenvs.2022.924842>.
- [13] J. Knisz, R. Eckert, L.M. Gieg, A. Koerd, J.S. Lee, E.R. Silva, T.L. Skovhus, B.A. An Stepec, S.A. Wade, Microbiologically influenced corrosion—more than just microorganisms, *FEMS Microbiol. Rev.* 47 (2023) fuad041, <https://doi.org/10.1093/femsre/fuad041>.
- [14] D. Rane, S. Kerkar, S.R. Ramanan, M. Kowshik, Superwetttable surfaces and factors impacting microbial adherence in microbiologically-influenced corrosion: a review, *World J. Microb. Biot.* 40 (2024) 98, <https://doi.org/10.1007/s11274-024-03886-3>.
- [15] L. Iannucci, M. Parvis, P. Cristiani, R. Ferrero, E. Angelini, S. Grassini, A novel approach for microbial corrosion assessment, *IEEE T. Instrum. Meas.* 68 (2019) 1424–1431, <https://doi.org/10.1109/TIM.2019.2905734>.
- [16] J. Zinke, E.D. Nilsson, P. Zieger, M.E. Salter, The effect of seawater salinity and seawater temperature on sea salt aerosol production, *J. Geophys. Res.—Atmos.* 127 (2022) e2021JD036005, <https://doi.org/10.1029/2021JD036005>.
- [17] W.A. Zoubi, Y.G. Ko, Self-assembly of hierarchical N-heterocycles-inorganic materials into three-dimensional structure for superior corrosion protection, *Chem. Eng. J.* 356 (2019) 850–856, <https://doi.org/10.1016/j.cej.2018.09.089>.
- [18] H. Tian, Z. Cui, H. Ma, P. Zhao, M. Yan, X. Wang, H. Cui, Corrosion evolution and stress corrosion cracking behavior of a low carbon bainite steel in the marine environments: effect of the marine zones, *Corros. Sci.* 206 (2022) 110490, <https://doi.org/10.1016/j.corsci.2022.110490>.
- [19] L. Chen, L.Y. Duan, M. Cui, R. Huang, R. Su, W. Qi, Z. He, Biomimetic surface coatings for marine antifouling: natural antifoulants, synthetic polymers and surface microtopography, *Sci. Total Environ.* 766 (2021) 144469, <https://doi.org/10.1016/j.scitotenv.2020.144469>.
- [20] H. Jin, J. Wang, L. Tian, M. Gao, J. Zhao, L. Ren, Recent advances in emerging integrated antifouling and anticorrosion coatings, *Mater. Des.* 213 (2022) 110307, <https://doi.org/10.1016/j.matdes.2021.110307>.
- [21] J. Sha, X. Liu, R. Chen, J. Yu, Q. Liu, J. Liu, J. Zhu, P. Liu, R. Li, J. Wang, Surface hydrolysis-eugenol self-polishing marine antifouling coating, *J. Colloid Interface Sci.* 637 (2023) 67–75, <https://doi.org/10.1016/j.jcis.2023.01.058>.
- [22] H. Zeng, F. Yin, L. Fan, C. Li, H. Lin, F. Liu, Q. Li, Combination of dexamethasone and dexmedetomidine as adjuvants of transversus abdominis plane block for postoperative analgesia in gastric cancer patients: a double-blinded randomized controlled trial, *J. Clin. Anesth.* 97 (2024) 111543, <https://doi.org/10.1016/j.jclinane.2024.111543>.
- [23] A.R. Bhalotra, M. Arya, R. Singh, S. Dhiman, Comparative evaluation of nebulized versus intravenous dexmedetomidine on intubating conditions during awake fiberoptic nasotracheal intubation, *J. Clin. Anesth.* 95 (2024) 111461, <https://doi.org/10.1016/j.jclinane.2024.111461>.
- [24] Y.Y. Cui, M.Q. Sheng, Y. Liu, Y. Feng, H.J. Li, Y.C. Wu, Antifouling and self-healing performance of marine coatings based on hydrogen-bond interactions, *ACS Appl. Mater. Interfaces* 15 (2023) 58967–58975, <https://doi.org/10.1021/acsami.3c16130>.
- [25] A.S. Krång, M. Dahlström, Effects of a candidate antifouling compound (medetomidine) on pheromone induced mate search in the amphipod corophium volutator, *Mar. Pollut. Bull.* 52 (2006) 1776–1783, <https://doi.org/10.1016/j.marpolbul.2006.09.015>.
- [26] A. Hilvarsson, C. Ohlsson, H. Blanck, A. Granmo, Bioaccumulation of the new antifouling medetomidine in marine organisms, *Mar. Environ. Res.* 68 (2009) 19–24, <https://doi.org/10.1016/j.marenvres.2009.03.007>.
- [27] L. Ulrika, A.R. Magnus, H.F. Linda, F. Susanna, B. Lars, M.L. Jonne, P. Katariina, V. Anne, P.K. Jyrki, G. Lina, S. Mika, G.E.M.L. Lena, B. Anders, Octopamine receptors from the barnacle balanus improvisus are activated by the alpha2-adrenoceptor agonist medetomidine, *Mol. Pharmacol.* 78 (2010) 237–248, <https://doi.org/10.1124/mol.110.063594>.
- [28] X. Li, X. Chen, J. Chen, D. Huo, X. Gao, J. Dong, Y. Yin, J. Liu, D. Nan, Development of an epoxy resin-based anticorrosive coating enhanced by monolayer amino-modified graphene oxide, *Prog. Org. Coat.* 194 (2024) 108624, <https://doi.org/10.1016/j.porgcoat.2024.108624>.
- [29] P.P. Vijayan, K. Formela, M.R. Saeb, P.G. Chithra, S. Thomas, Integration of antifouling properties into epoxy coatings: a review, *J. Coat. Technol. Res.* 19 (2021) 269–284, <https://doi.org/10.1007/s11998-021-00555-0>.
- [30] M.S. Noor Idora, M. Ferry, W.B. Wan Nik, S. Jasnizat, Evaluation of tannin from rhizophora apiculata as natural antifouling agents in epoxy paint for marine application, *Prog. Org. Coat.* 81 (2015) 125–131, <https://doi.org/10.1016/j.porgcoat.2014.12.012>.
- [31] A.Y. Musa, A.A.H. Kadhum, A.B. Mohamad, M.S. Takriff, Experimental and theoretical study on the inhibition performance of triazole compounds for mild steel corrosion, *Corros. Sci.* 52 (2010) 3331–3340, <https://doi.org/10.1016/j.corsci.2010.06.002>.
- [32] M. Li, Y.Y. Ouyang, W.Z. Yang, Y. Chen, K.G. Zhang, Z.S. Zuo, X.S. Yin, Y. Liu, Inhibition performances of imidazole derivatives with increasing fluorine atom contents in anions against carbon steel corrosion in 1 M HCl, *J. Mol. Liq.* 322 (2021) 114535, <https://doi.org/10.1016/j.molliq.2020.114535>.
- [33] S.K. Saha, A. Hens, N.C. Murmu, P. Banerjee, A comparative density functional theory and molecular dynamics simulation studies of the corrosion inhibitory action of two novel N-heterocyclic organic compounds along with a few others over steel surface, *J. Mol. Liq.* 215 (2016) 486–495, <https://doi.org/10.1016/j.molliq.2016.01.024>.
- [34] D. Yang, Y. Ye, Y. Su, S. Liu, D. Gong, H. Zhao, Functionalization of citric acid-based carbon dots by imidazole toward novel green corrosion inhibitor for carbon steel, *J. Clean. Prod.* 229 (2019) 180–192, <https://doi.org/10.1016/j.jclepro.2019.05.030>.
- [35] W. Zhang, H.J. Li, Y. Li, C. Ma, Q. Pan, L. Chen, L.J. Wang, Y.C. Wu, Anticorrosion performance of acriflavine-Zn<sup>2+</sup> system for mild steel in seawater utilization, *J. Mol. Liq.* 299 (2020) 112152, <https://doi.org/10.1016/j.molliq.2019.112152>.
- [36] W. Zhang, L. Chen, Y.C. Wu, S. Chen, Y. Gao, L.J. Wang, Y. Luo, Q. Li, H.J. Li, Preparation of methylacridinium iodides self-assembled monolayers and its anti-corrosion properties for mild steel in seawater: experimental and computational studies, *J. Mol. Liq.* 313 (2020) 113545, <https://doi.org/10.1016/j.molliq.2020.113545>.
- [37] W. Zhang, H.J. Li, L. Chen, J. Sun, X. Ma, Y. Li, C. Liu, X. Han, B. Pang, Y.C. Wu, Performance and mechanism of a composite scaling–corrosion inhibitor used in seawater: 10-Methylacridinium iodide and sodium citrate, *Desalination* 486 (2020) 114482, <https://doi.org/10.1016/j.desal.2020.114482>.
- [38] D.Y. Wang, H.J. Li, X. Chen, B.L. Nie, Y.C. Wu, Fabricating of grape seed proanthocyanidins loaded Zein-NaCas composite nanoparticles to exert effective

- inhibition of Q235 steel corrosion in seawater, *J. Mol. Liq.* 348 (2022) 118467, <https://doi.org/10.1016/j.molliq.2022.118467>.
- [39] X.F. Yang, F. Li, W.W. Zhang, 4-(Pyridin-4-yl) thiazol-2-amine as an efficient non-toxic inhibitor for mild steel in hydrochloric acid solutions, *RSC Adv.* 9 (2019) 10454–10464, <https://doi.org/10.1039/c8ra09618j>.
- [40] M. Zhang, F. Xu, D. Lin, J. Peng, Y. Zhu, H. Wang, A smart anti-corrosion coating based on triple functional fillers, *Chem. Eng. J.* 446 (2022) 137078, <https://doi.org/10.1016/j.cej.2022.137078>.